

1. Fie φ vect euclidian $E_3 = (\mathbb{R}^3 / \varphi, \langle \cdot, \cdot \rangle)$

$$B_0 = \{e_1, e_2, e_3\} \subset E_3$$

$$T: E_3 \rightarrow E_3$$

$$T(e_1) = \frac{2}{3}e_1 + \frac{2}{3}e_2 - \frac{1}{3}e_3$$

$$T(e_2) = \frac{2}{3}e_1 - \frac{1}{3}e_2 + \frac{2}{3}e_3$$

$$T(e_3) = -\frac{1}{3}e_1 + \frac{2}{3}e_2 + \frac{2}{3}e_3$$

$$A = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$A^t = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$A \cdot A^t = I_3 \Rightarrow A \text{ m ortogonală} \Rightarrow$$

$\Rightarrow T$ este transp ortogonală

2. $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$

$$Q(x) = 3x_1^2 + 4x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_2x_3 \quad (b) \quad x = (x_1, x_2, x_3) \in \mathbb{R}^3$$

a) metoda Gauss

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 4 & -2 \\ 0 & -2 & 5 \end{pmatrix}$$

$$Q(x) = 3\left(x_1 + \frac{2}{3}x_2\right)^2 + \frac{4}{3}x_2^2 + 4x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_2x_3$$

$$= 3\left(x_1 + \frac{2}{3}x_2\right)^2 + \frac{4}{3}\left(x_2 - \frac{3}{4}x_3\right)^2 - \frac{3}{2}x_3^2 + 5x_3^2$$

$$= 3\left(x_1 + \frac{2}{3}x_2\right)^2 + \frac{4}{3}\left(x_2 - \frac{3}{4}x_3\right)^2 + \frac{7}{2}x_3^2$$

och de word :

$$\begin{cases} x_1' = x_1 + \frac{2}{3} x_2 \\ x_2' = x_2 - \frac{3}{4} x_3 \\ x_3' = x_3 \end{cases}$$

$$Q(x') = 3(x_1')^2 + \frac{9}{3}(x_2')^2 + \frac{7}{2}(x_3')^2 \quad (b) \quad x' = (x_1', x_2', x_3') \in \mathbb{R}^3$$

$$A' = \begin{pmatrix} 3 & 0 & 0 \\ 0 & \frac{9}{3} & 0 \\ 0 & 0 & \frac{7}{2} \end{pmatrix} \quad \text{— matrix lin } Q \text{ in basis } B'$$

$$B' = \{ e_1' = (1, 0, 0), e_2' = (\frac{2}{3}, 1, 0), e_3' = (0, -\frac{3}{4}, 1) \}$$

h)

$$\begin{cases} \Delta_1 = 3 \neq 0 \\ \Delta_2 = 9 \neq 0 \\ \Delta_3 = 28 \neq 0 \end{cases} \Rightarrow \Delta_i \neq 0, (b) \quad i = \overline{1, 3}$$

$$\Rightarrow Q(x) = \frac{1}{3}(x_1')^2 + \frac{3}{9}(x_2')^2 + \frac{2}{7}(x_3')^2 \quad (b) \quad x = (x_1', x_2', x_3') \in \mathbb{R}^3$$

1)

$$P(\lambda) = \begin{vmatrix} 3-\lambda & 2 & 0 \\ 2 & 4-\lambda & -2 \\ 0 & -2 & 5-\lambda \end{vmatrix} = (\lambda-1)(\lambda-4)(\lambda-7)$$

$\begin{matrix} \lambda_1 = 1 \\ \lambda_2 = 4 \\ \lambda_3 = 7 \end{matrix}$

$$\S_{\lambda_1=1} \begin{cases} 2x + 2y = 0 \\ 2x + 3y - 2z = 0 \\ -2y + 4z = 0 \end{cases}$$

$$\Delta_{\lambda_1} = \begin{vmatrix} 2 & 2 \\ 2 & 3 \end{vmatrix} = 2 \neq 0 \Rightarrow y = 2$$

$$\begin{cases} x, y \text{ nec } m \\ z = x \text{ nec } n \end{cases} \Rightarrow \begin{matrix} z = 2x \\ x = -2x \end{matrix}$$

$$V_{\lambda_1} = \{ x \cdot \frac{v_1}{\sqrt{2}} \mid x \in \mathbb{R} \} = \{ x \cdot v_1 \mid x \in \mathbb{R} \}$$

$$S_{\lambda_2} = 4 \begin{cases} -4x + 2y = 0 \\ 2x - 2z = 0 \\ -2y + z = 0 \end{cases}$$

$$\Delta \lambda = \begin{vmatrix} -4 & 2 \\ 2 & 0 \end{vmatrix} = -4 \neq 0 \Rightarrow xy = 2$$

$$\begin{cases} x, y \text{ nec nec} \\ z = p \text{ nec nec} \end{cases} \Rightarrow \begin{cases} x = p \\ y = \frac{p}{2} \end{cases}$$

$$v_{\lambda_2} = \left\{ p \left(1, \frac{1}{2}, 1 \right) / p \in \mathbb{R} \right\} = \{ p v_2 / p \in \mathbb{R} \}$$

$$S_{\lambda_3} = 7 \begin{cases} -4x + 2y = 0 \\ 2x - 3y - 2z = 0 \\ -2y - 2z = 0 \end{cases}$$

$$\Delta \lambda = \begin{vmatrix} -4 & 2 \\ 2 & -3 \end{vmatrix} = 8 \neq 0 \Rightarrow xy = 2$$

$$\begin{cases} x, y \text{ nec nec} \\ z = p \text{ nec nec} \end{cases} \Rightarrow \begin{cases} y = -x \\ x = -\frac{p}{2} \end{cases}$$

$$v_{\lambda_3} = \left\{ p \left(-\frac{1}{2}, -1, 1 \right) / p \in \mathbb{R} \right\} = \{ p v_3 / p \in \mathbb{R} \}$$

$$B = \{ v_1, v_2, v_3 \}$$

bază ortonormală (i.e. $\langle v_i, v_j \rangle = 0$)

$$B' = \left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \frac{v_3}{\|v_3\|} \right\} = \frac{1}{3} (-2, 2, 1), \frac{1}{3} (2, 1, 2), \frac{1}{3} (-1, -2, 2)$$

bază ortonormală

$$Q(x) = (x_1')^2 + 4(x_2')^2 + 7(x_3')^2, \quad (x) = (x_1', x_2', x_3') \in \mathbb{R}^3$$

$$3. \quad \Gamma: x_1^2 + 2x_1x_2 - 5x_2^2 + 4x_1 - 8x_2 - 17 = 0$$

- forma canonică a conicei Γ prin isometrie.

$$A = \begin{pmatrix} 1 & +1 \\ +1 & -5 \end{pmatrix}; \quad \delta = \det A = \cancel{0} - 6$$

$$A' = \begin{pmatrix} 1 & +1 & 2 \\ +1 & -5 & -4 \\ 2 & -4 & -17 \end{pmatrix}; \quad \Delta = \det A' = \cancel{12} 90$$

$$\delta < 0 \quad \Delta \neq 0 \quad \Rightarrow \quad \Gamma \text{ este hiperbolă}$$

- centrul conicei Γ , $P_0(x_1^0, x_2^0)$

$$\begin{cases} \frac{\partial \Gamma}{\partial x_1} = 0 \\ \frac{\partial \Gamma}{\partial x_2} = 0 \end{cases} \Leftrightarrow \begin{cases} 2x_1 + 2x_2 + 4 = 0 \\ -10x_2 + 2x_1 - 8 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \\ x_2 = -1 \end{cases}$$

~~$P_0(-\frac{4}{9}, -\frac{16}{9})$ centrul conicei Γ~~ $P_0(-1, -1)$ centrul conicei Γ

- translația t :

$$t \begin{cases} x_1' = x_1 - x_1^0 \\ x_2' = x_2 - x_2^0 \end{cases} \Leftrightarrow \begin{cases} x_1' = x_1 + \frac{4}{9} \\ x_2' = x_2 + \frac{16}{9} \end{cases} \Leftrightarrow \begin{cases} x_1 = x_1' - \frac{4}{9} \\ x_2 = x_2' - \frac{16}{9} \end{cases}$$

$$t(\Gamma): \cancel{(x_1')^2 - 5(x_2')^2}$$

$$t \begin{cases} x_1' = x_1 - x_1^0 \\ x_2' = x_2 - x_2^0 \end{cases} \Leftrightarrow \begin{cases} x_1' = x_1 + 1 \\ x_2' = x_2 + 1 \end{cases} \Leftrightarrow \begin{cases} x_1 = x_1' - 1 \\ x_2 = x_2' - 1 \end{cases}$$

$$t(\Gamma): (x_1')^2 - 5(x_2')^2 + 2x_1'x_2' - 2(-1-1+2)x_1' - 2(-1+5-4)x_2' - 15 = 0$$

$$t(\Gamma): (x_1')^2 + 2x_1'x_2' - 5(x_2')^2 - 15 = 0$$

$$p(\lambda) = \begin{vmatrix} 1 - \lambda & 1 \\ 1 & -5 - \lambda \end{vmatrix} = \lambda^2 + 4\lambda - 6$$

$$p(\lambda) = 0 \Rightarrow \lambda^2 + 4\lambda - 6 = 0 \quad \begin{cases} \lambda_1 = -2 - \sqrt{10} \\ \lambda_2 = -2 + \sqrt{10} \end{cases}$$

$$S'_{\lambda_1} \begin{cases} (3 + \sqrt{10})x + y = 0 \\ x + (-3 + \sqrt{10})y = 0 \end{cases} \Rightarrow y = (3 - \sqrt{10})x$$

$$\Delta x = 3 + \sqrt{10} \quad y = 1$$

$$y = x \Rightarrow x = (3 - \sqrt{10})x$$

$$V_{\lambda_1} = \{ \alpha (3 - \sqrt{10}, 1) / \alpha \in \mathbb{R} \}$$

$$S'_{\lambda_2} \begin{cases} (3 - \sqrt{10})x + y = 0 \\ x + (-3 - \sqrt{10})y = 0 \end{cases}$$

$$\Delta x = 3 - \sqrt{10} \quad y = 1$$

$$y = \beta \Rightarrow x = (3 + \sqrt{10})\beta$$

$$V_{\lambda_2} = \{ \beta (3 + \sqrt{10}, 1) / \beta \in \mathbb{R} \}$$

— basis vect proper:

$$\begin{cases} x_1 = (3 - \sqrt{10}, 1) \\ x_2 = (3 + \sqrt{10}, 1) \end{cases}$$

$$\langle x_1, x_2 \rangle = 0 \Rightarrow x_1 \perp x_2$$

— Normalized vect x_1, x_2 :

$$\begin{cases} e_1 = \frac{x_1}{\|x_1\|} \\ e_2 = \frac{x_2}{\|x_2\|} \end{cases} \Rightarrow \begin{cases} \frac{1}{\sqrt{20 - 6\sqrt{10}}} (3 - \sqrt{10}, 1) \\ \frac{1}{\sqrt{20 + 6\sqrt{10}}} (3 + \sqrt{10}, 1) \end{cases}$$

efectuăm rotația n :

$$n \begin{cases} x_1'' = \frac{1}{\sqrt{20-6\sqrt{10}}} ((3-\sqrt{10})x_1' + x_2') \\ x_2'' = \frac{1}{\sqrt{20+6\sqrt{10}}} ((3+\sqrt{10})x_1' + x_2') \end{cases}$$

$$R = \begin{pmatrix} \frac{3-\sqrt{10}}{\sqrt{20-6\sqrt{10}}} & \frac{1}{\sqrt{20-6\sqrt{10}}} \\ \frac{3+\sqrt{10}}{\sqrt{20+6\sqrt{10}}} & \frac{1}{\sqrt{20+6\sqrt{10}}} \end{pmatrix}$$

$$R^T = \begin{pmatrix} \frac{3-\sqrt{10}}{\sqrt{20-6\sqrt{10}}} & \frac{3+\sqrt{10}}{\sqrt{20+6\sqrt{10}}} \\ \frac{1}{\sqrt{20-6\sqrt{10}}} & \frac{1}{\sqrt{20+6\sqrt{10}}} \end{pmatrix}$$

$$R \cdot R^T = \begin{pmatrix} \frac{20-6\sqrt{10}}{20-6\sqrt{10}} & 0 \\ 0 & \frac{20-6\sqrt{10}}{20+6\sqrt{10}} \end{pmatrix} = I_2 \Rightarrow R \text{ matrice ortogonală}$$

$$n^{-1} = R^T \Rightarrow \begin{cases} x_1' = \frac{3-\sqrt{10}}{\sqrt{20-6\sqrt{10}}} x_1'' + \frac{3+\sqrt{10}}{\sqrt{20+6\sqrt{10}}} x_2'' \\ x_2' = \frac{1}{\sqrt{20-6\sqrt{10}}} x_1'' + \frac{1}{\sqrt{20+6\sqrt{10}}} x_2'' \end{cases}$$

$$(rot)(\Pi): (-2-\sqrt{10})(x_1'')^2 + (-2+\sqrt{10})(x_2'')^2 - 15 = 0 \quad / : +15$$

~~formă canonică a~~

$$-\frac{(x_1'')^2}{\frac{15}{2+\sqrt{10}}} + \frac{(x_2'')^2}{\frac{15}{-2+\sqrt{10}}} - 1 = 0$$

$$-\frac{(x_1'')^2}{a^2} + \frac{(x_2'')^2}{b^2} - 1 = 0,$$

$$\text{unde } a = \sqrt{\frac{15}{2+\sqrt{10}}} \text{ și } b = \sqrt{\frac{15}{-2+\sqrt{10}}}$$

forma canonică a conicei Π obț prin izometrie